Emotional intensity

Bookmark this page

XPERT

Problem: Nothing can be further from emotions than the cold computations of AIT!

And yet, Algorithmic Information is crucial to compute emotions. Not the emotions themselves, but their intensity. Emotions elicited by the occurrence of events may be intense. Their intensity is due to two factors:

1. The type of event (are there casualties, or are we talking of a 10€ loss);

2. The event's unexpectedness.

Simplicity is of course at work in **unexpectedness**, since unexpected situations are by definition abnormally simple. As we saw when we discussed the "law of distant deaths", **simplicity** is involved in the amplitude of the emotional stimulus. For instance, under the hypothesis that the impact of each casualty is proportional to its complexity, we can derive that emotional intensity varies as the **logarithm of their number**.

Wheel of Fortune 1

0/1 point (ungraded)

In a wheel of fortune game, the player launches the wheel which then stops at one of the N discrete spaces of the wheel. There is only one winning space. Suppose that the player has won.

What is the contribution of N to her emotional intensity?

- ( ) $log_2\left(N\right)$ ✔
- ( ) $log_2\left(2 \times N\right)$
- (•) $2 \times log_2\left(N\right)$
- ( ) N
- ( ) $2 \times N$

Answer
Incorrect:
We compute unexpectedness. Causal complexity is $C_w = log_2\left(N\right)$ bits. Description complexity is $C_d = 0$, as the winning space is pre-determined. So the correct answer is $log_2\left(N\right)$.

Submit You have used 1 of 1 attempt

Answers are displayed within the problem

Imagine for a moment that the player, though having won, is denied the reward. The value of emotional intensity would be the same as for winning, though with an opposite sign.

Let's suppose now that the player did not win because she missed the winning space. She might be upset as well.
In these kinds of situations, people tend to consider the unexpectedness of the **counterfactual** win. They are all the more upset as they failed by a small margin.
Their (negative) emotion is, in absolute value, as intense as the emotion of winning, minus the counterfactual unexpectedness of reaching the winning space.

Wheel of Fortune 2

1/1 point (ungraded)

Suppose that the wheel stopped at distance k of the winning space (near miss).

What is the contribution of k to the absolute value of the (now negative) emotional intensity?

- [x] $-(1 + log_2(k))$
- [ ] $-(1 + 2 \times log_2(k))$

✔

Answer
Correct:
Since the description complexity of the winning space is 0, we just need to consider the causal complexity of reaching the winning space from the landing space, *i.e.* the minimal changes in the "world" needed to reach it. If we take the landing sector as origin, the "world" needs 1 bit for the direction plus $log_2(k)$ for the intensity of the force.

Submit You have used 1 of 1 attempt

✔ Correct (1/1 point)

‹ Previous Next ›